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Course: Artificial Intelligence

Program 3: Water Jug Problem

Aim: To implement the problem in Python.

Procedure:

1. Start.
2. Define initial state.
3. Apply search strategy.
4. Find solution.
5. Display result.
6. Stop.

Algorithm:

1.Start 2.Initialize 3.Generate states 4.Check goal 5.Display solution 6.Stop.

Python Program:

```
from collections import deque
def water_jug(a,b,target):
    q=deque([(0,0)]);vis={(0,0)}
    while q:
        x,y=q.popleft()
        if x==target or y==target:return True
    return False
```

Observation: The program successfully finds a valid solution.

Result: The Python program was executed successfully.

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Program 4: Crypt-Arithmetic Problem

Aim: To implement the problem in Python.

Procedure:

1. Start.
2. Define initial state.
3. Apply search strategy.
4. Find solution.
5. Display result.
6. Stop.

Algorithm:

1.Start 2.Initialize 3.Generate states 4.Check goal 5.Display solution 6.Stop.

Python Program:

```
import itertools
letters='SENDMORY'
# Crypt-arithmetic solver outline
```

Observation: The program successfully finds a valid solution.

Result: The Python program was executed successfully.